Induction & Recursion

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1 Theorems about transifinite- and epsilon- induction and recursion.

Recall 1.1. Transfinite recursion
$G : V \to V$
$\exists! F : \mathrm{Ord} \to V$, s.t.
$F(a) = G(F \restriction a)$

Recall 1.2. Epsilon recursion:
$G : V \to V$
$\exists! F : V \to V$, s.t.
$F(a) = G(F \restriction a)$

Definition 1.1. Height function definition:
$h(x)$ is the height of a the set. It is equal by definition to the maximum length of a $\ni$-chain, started from x.

Examples:
$h(\emptyset)$=0,
$h(\{\emptyset\})$=1,
$h(\{\emptyset, \{\emptyset\}\})$=2,
and in general if n is a finite ordinal number, then $h(n) = n$.
Definition: by $\in$-recursion

$$h(x) := \sup\{h(y) | y \in x\}$$

Property:

$$h(x) = \sup h[x]$$

$G(R) := \sup(\mathrm{ran}(R)) = \bigcup(\mathrm{ran}(R))$
via epsilon recursion.
$h : V \to V$
$h(a) = \sup(\mathrm{ran}(h \restriction a))$

Lemma 1.1. ()

$$(h : V \to \mathrm{Ord})$$

Proof.
by epsilon induction.
Done.

1.1 older Inductions and recursions thms

There exist unique class-function rec, such that:

$$\mathrm{rec} : (V \times V \to V) \to (V \to V)$$

$$\mathrm{rec}(H) = H \ \circ \ (\mathrm{rec}(H) \restriction_{_})$$